EnergyPlus Has More

Q: I'm an energy-conscious architect who has been using software simulations in my work designing homes and doing home renovations. However, I'm not satisfied with the existing software programs that I'm aware of. For example, I want a simulation program with more flexibility in matching the simulation to the actual residential building system configurations. Can you point me to any new software that makes up for some of the weaknesses of the existing programs?

—Seeking Superb Simulations

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A: In 1996 the U.S. Department of Energy (DOE) began developing a new building energy simulation program called EnergyPlus. Released in April 2001, EnergyPlus is an all-new program, building on the most popular features and capabilities of BLAST and DOE-2, two previously created building systems simulation software programs (see "Putting the Byte into Your Analysis Toolkit," *HE* Sept/Oct '98, p. 25). The EnergyPlus development team included the U.S. Army Construction Engineering Research Laboratory, Lawrence Berkeley National Laboratory, University of Illinois at Champaign-Urbana, Florida Solar Energy Center, Oklahoma State University, GARD Analytics, and DOE.

BLAST is based on the National Bureau of Standards Heating and Cooling Load Determination program (NBSLD), developed at the National Bureau of Standards (now NIST) in the early 1970s. DOE–2 was developed by Lawrence Berkeley National Laboratory and many other organizations, funded primarily by DOE. DOE-2 is based on the Post Office program written in the late 1960s. The main difference between the programs is their load calculation method—DOE–2 uses a room-weighting factor approach, while BLAST uses a heat balance approach.

When DOE-2 was originally developed, computational time was expensive. The room weighting factor method was much faster than the heat balance method. Today's PCs are many times faster than the mainframe computers of 20 or more years ago. EnergyPlus allows users to evaluate realistic system controls, moisture adsorption and desorption in building elements, radiant heating and cooling systems, and interzone air flow—none of which DOE-2 or BLAST could simulate well.

EnergyPlus uses an integrated simulation solution. This means that loads calculated at a user-specified time step (15-minute default) are passed to the building systems simulation module at the same time. The building systems simulation module, with a variable time step (down to seconds), calculates the response of the building's heating, cooling, and electrical systems. If the building systems cannot meet the loads, the space temperatures are adjusted at the next load calculation time step. This integrated solution technique overcomes one deficiency of both BLAST and DOE–2—inaccurate space temperature prediction due to no feedback on unmet loads (caused by undersized HVAC systems, for example).

Heat and Mass Balance

The heat balance calculations used in EnergyPlus are derived from IBLAST, a research version of BLAST that integrated the HVAC systems and building loads simulation. Three popular existing capabilities from DOE-2 were added to EnergyPlus: daylighting (see Figure 1), advanced fenestration, and nonuniform distribution of solar radiation and illumination from the sun, known as anisotropic sky. All of the existing DOE-2 daylighting calculations were included in EnergyPlus, while future daylighting updates will include interior interreflection calculation, reflection from neighboring buildings, and the handling of complex fenestration such as blinds, light shelves, and roof monitors.

The latest accurate fenestration algorithms calculate the angular dependence of transmission and absorption for solar and visible radiation, temperature-dependent U-value, coatings, and framing elements. Movable interior and exterior window shades and electrochromic glazing can now also be simulated. The anisotropic sky model improves the calculation of diffuse and direct solar radiation and illumination.

The major enhancement of the EnergyPlus heat balance engine over BLAST is that EnergyPlus includes mass transfer and radiant heating and cooling in its modeling. EnergyPlus calculates layer-by-layer mass transfer through surfaces and the zone air mass balance. Radiant heating and cooling models allow more accurate thermal comfort calculations

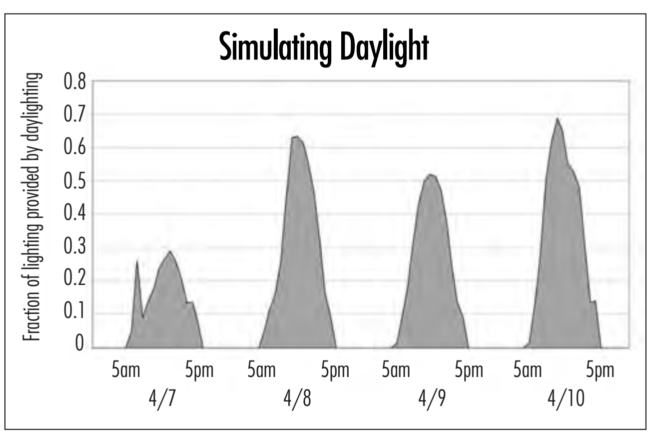

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Figure 1. EnergyPlus includes daylighting capabilities from the DOE-2 software program, allowing users to simulate how natural daylight will affect the interior of a building.

within EnergyPlus along with improved control modeling.

Building Systems

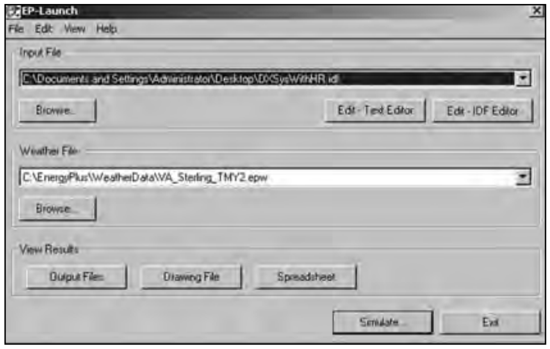
In EnergyPlus, the fixed system types of DOE–2 and BLAST (such as constant-volume reheat, or VAV) are replaced by heating and cooling equipment components that users can configure to more closely match their system configurations. For each major system type in BLAST and DOE-2, we created an equivalent system file—a subset of an input file—with all the required heating and cooling components. Users can modify these system templates to match their building and system configuration.

In EnergyPlus, air and water loops mimic the network of ducts and pipes found in real buildings. Air loops simulate air movement, conditioning, and mixing, and include fans, coils, heat recovery, supply air temperature, and outside air economizer controls. Zone equipment—such as reheat coils, mixing and other dampers, local convection units, baseboard, radiators, and low-temperature radiant panels—connect the air loop to the zone. An important EnergyPlus feature is the ability to specify more than one zone equipment type for each zone. For the water loops, users select and specify equipment by type and by the equipment's relevant operating characteristics. A similar electrical loop for simulating electrical systems is planned for Version 1.1—simulating supply (utility, PV modules, and fuel cells) and demand (plug loads, lighting, and other loads).

Input, Output, and Weather Data

EnergyPlus input, output, and weather data files are ASCII text files with a simpler structure than data files in either DOE–2 or BLAST. We designed these data files to make it easier to get building data from, and share them with, other sources, such as CAD programs and user interfaces. EnergyPlus input and output data files were never intended to be the main interface for typical end users. (DOE–2 has more than 15 private sector interfaces.) Several user interface developers began creating interfaces for EnergyPlus in the spring of 2001, which we expect to be available in late 2001. Indeed, connectivity and extensibility were overriding objectives when we designed and developed EnergyPlus, to facilitate the possibility of users writing new modules and interfaces for EnergyPlus in the future.

One further feature of EnergyPlus is a utility that converts geometry data from CAD drawings, using the International Alliance for Interoperability's Industry Foundation Classes (IFCs), to EnergyPlus input data files. The weather data



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EnergyPlus allows users to access and input data from a variety of sources, including the TMY2 weather data.

include much of the same data as the TMY2 weather data set, but EnergyPlus allows and reads subsets of years and even subhourly data. The weather data format includes a "minutes" field that allows users to include recorded data for shorter time periods (1, 5, and 15 minutes) when they are available. Where the user has selected a simulation time step shorter than the weather data time step, EnergyPlus just interpolates. Most users will use hourly data such as TMY2.

How Accurate Is EnergyPlus?

While developing EnergyPlus, we continually validated the results, using several test methods. We focused primarily on comparative and analytical testing such as that described in ASHRAE Standard 140-2001—a basic rectangular-shaped room with a series of tests including windows and shading for both low-mass and high-mass construction. Testing EnergyPlus against itself has been extremely useful in detecting and resolving problems introduced during development. The testing to date demonstrates that EnergyPlus provides results in good agreement with other simulation programs for simple cases. Testing results are available on the EnergyPlus Web site.

Version 1.0 and Beyond

DOE released EnergyPlus Version 1.0 in April 2001 and plans to release updates to EnergyPlus on a regular schedule. New features already under development are electrical-system simulation, fuel cells, advanced fenestration and daylighting, and other building technologies. These and other features will be included in the second major release (1.1), currently planned for 2003.

For more information:

Up-to-date information on EnergyPlus, including availability of updates, licensing, weather data, programming standards, and other documentation, is available on the EnergyPlus Web site:

www.eren.doe.gov/buildings/energy_tools/energyplus.

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